Executive Summary: Customer Churn Analysis

This analysis aims to identify the primary drivers of customer churn and provide actionable insights to improve customer retention. The study is based on a comprehensive exploratory data analysis supported by well-structured visualizations, enabling clear interpretation of churn behavior across demographics, tenure, services, and payment methods.

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1. Data Preparation & Validation

- The dataset was examined for missing values, incorrect data types, and duplicates.
- The **TotalCharges** column contained blank entries, which were found to correspond to customers with **0 months of tenure**. These were logically replaced with `0`, and the column was converted to a numeric format.
- The **SeniorCitizen** variable was transformed from binary values (0/1) into **Yes/No** for better interpretability.
- After cleaning, the dataset was confirmed to be consistent and analysis-ready.

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2. Overall Churn Overview

- Approximately **26–27% of customers have churned**, while **73–74% remain active**.
- This indicates that **nearly 1 out of every 4 customers leaves the service**, highlighting churn as a significant business concern.

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3. Demographic Insights

Gender

- Churn is **almost evenly distributed across genders**.

- Both male and female customers show churn rates close to the overall average (~26%), indicating that **gender is not a strong churn predictor**.

Senior Citizens

- **Senior citizens account for a disproportionately higher churn rate**.
- While they represent a smaller portion of the customer base, **over 40% of senior citizens churn**, compared to **around 23–25% among non-senior customers**.
- This suggests the need for tailored engagement strategies for elderly customers.

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4. Tenure-Based Churn Behavior

- Churn is **heavily front-loaded**:
  - **More than 50% of all churned customers leave within the first 1–2 months**.
  - The **highest churn rate occurs at month 1**, after which churn declines sharply.
- Customers who stay beyond **6 months show significantly higher retention**, and churn becomes almost negligible after long tenure.
- This highlights the **critical importance of early customer onboarding and experience**.

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5. Contract Type Analysis

- **Month-to-month contracts show the highest churn**, accounting for **over 85–90% of total churned customers**.
- Customers on **one-year and two-year contracts churn at rates below 10%**, demonstrating much stronger loyalty.
- Long-term contracts clearly act as a **churn-reducing mechanism**.

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6. Service-Level Insights

Internet Service

- Customers using **Fiber Optic internet show the highest churn rate**, with **over 40% churning**.
- DSL users and customers without internet services exhibit substantially lower churn.

Add-on Services (Security & Support)

Customers **without value-added services** consistently show higher churn:

- **Online Security**: Customers without it churn at **~40%**, compared to **~15% with it**
- **Tech Support**: Churn drops by **nearly 50%** when tech support is present
- **Online Backup & Device Protection** show similar patterns

These findings indicate that **service bundling significantly improves retention**.

Entertainment Services

- **Streaming TV and Streaming Movies** do not show a strong or consistent impact on churn.
- Customers churn at similar rates regardless of these services, suggesting they are **not primary churn drivers**.

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7. Payment Method Insights

- **Electronic Check users show the highest churn rate**, accounting for **~45% churn** within this group.

- Other payment methods (credit card, bank transfer, mailed check) demonstrate **much lower churn rates**, typically below **20%**.
- This may indicate trust, convenience, or billing experience issues with electronic checks.

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8. Key Business Recommendations

1. **Strengthen early engagement**
    - Focus retention efforts within the **first 1–3 months**, where churn risk is highest.
2. **Promote long-term contracts**
    - Incentivize customers to move from month-to-month to annual plans.
3. **Bundle security and support services**
    - Offer discounts or default bundles for Online Security, Tech Support, and Backup services.
4. **Target high-risk segments**
    - Senior citizens, fiber optic users, and electronic check payers require proactive engagement.
5. **Improve onboarding for fiber users**
    - High churn despite premium pricing suggests unmet expectations or service issues.

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9. Conclusion

The analysis clearly shows that churn is **not random**, but driven by tenure, contract type, service adoption, and payment behavior. By focusing on early-stage customers, promoting bundled services, and encouraging long-term commitments, the business can significantly reduce churn and improve customer lifetime value.